

- 5 (a) The resistance of a resistor is the ratio of potential difference across the resistor to the current in it.

(b) (i) $V_X = E - Ir$

From graph of V against I , gradient $= -r$

$$r = -\text{gradient} = -\left(\frac{5.40 - 4.20}{0.40 - 1.20}\right)$$
$$= 1.5 \, \Omega$$

OR

Using substitution of point

$$4.80 = 6.0 - 0.80 r$$

$$r = 1.5 \, \Omega$$

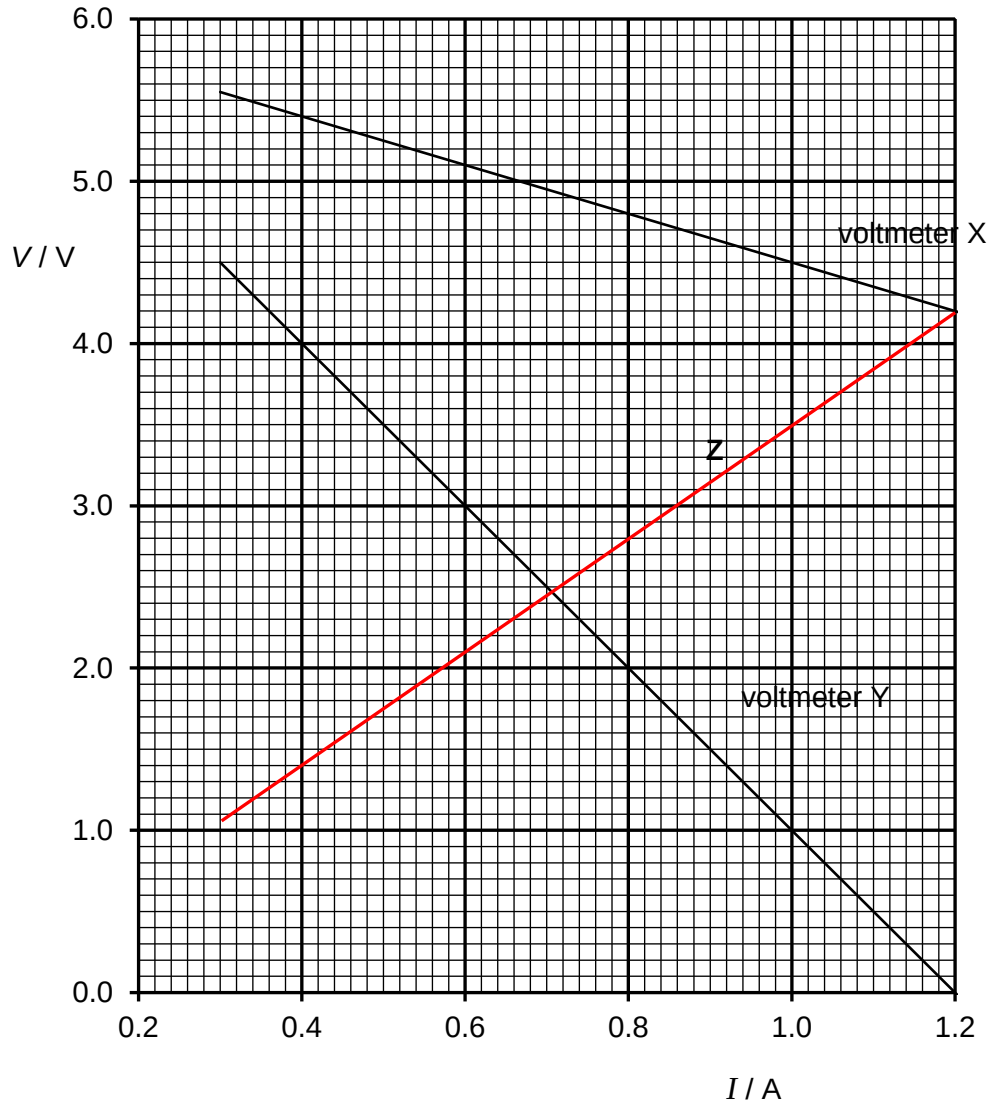
- (ii) When $I = 0.40 \, \text{A}$, p.d. across fixed resistor $= V_X - V_Y = 5.40 - 4.00 = 1.40 \, \text{V}$

$$R_s = \frac{p.d.}{I} = \frac{1.40}{0.40} = 3.5 \, \Omega \quad (\text{shown})$$

- (c) There is a maximum value of the resistance in the circuit, and from the given electromotive force of $6.0 \, \text{V}$, there will be a minimum current from $I = \frac{e.m.f.}{R}$.

- (d) When $V_X = 5.25 \, \text{V}$, $I = 0.50 \, \text{A}$. $V_Y = 3.50 \, \text{V}$
 $P = IV = (0.50)(3.50) = 1.75 \, \text{W}$

(e) (i)



(ii) 0.70 A (intersection of graph Z and voltmeter Y)

6 (c) The charge is stationary.